

DIGIMEDS : An Intelligent System for Digitizing Handwritten Medical Prescriptions

A major project report submitted in partial fulfilment of the requirement for the
award of degree of

Bachelor of Technology
in
Computer Science & Engineering

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DECLARATION

We hereby declare that the work presented in this major project report entitled 'DIGIMEDS : An Intelligent System for Digitizing Handwritten Medical Prescriptions' submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering, in the Department of Computer Science & Engineering and Information Technology, Jaypee University of Information Technology, Waknaghat, is an authentic record of our own work carried out during the period from January 2026 to may 2026 under the supervision of **Mr. Prateek Thakral** .We further declare that the matter embodied in this report has not been submitted for the award of any other degree or diploma at any other university or institution.

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SUPERVISOR'S CERTIFICATE

This is to certify that the major project report entitled '**DIGIMEDS: An Intelligent System for Digitizing Handwritten Medical Prescriptions**', submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science & Engineering**, in the Department of Computer Science & Engineering and Information Technology, Jaypee University of Information Technology, Waknaghat, is a bonafide project work carried out under my supervision during the period from **January 2026 to May 2026**. I have personally supervised the research work and confirm that it meets the standards required for submission. The project work has been conducted in accordance with ethical guidelines, and the matter embodied in the report has not been submitted elsewhere for the award of any other degree or diploma.

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TABLE OF CONTENT

Title		Page No.
List of Tables		II
List of Figures		III
List of Abbreviations, Symbols or Nomenclature		IV
Abstract		V
Chapter-1 (Introduction)		6-10
1.1	Introduction	
1.2	Problem Statement	
1.3	Objectives	
1.4	Significance and Motivation of the Project Work	
1.5	Organization of the report	
Chapter-2 (Literature Survey)		11-24
2.1	Overview of Relevant Literature	
2.2	Key Gaps in the Literature	
Chapter-3 (System Development)		25-35
3.1	Requirements and Analysis	
3.2	Project Design and	
3.3	Architecture Data Preparation	
3.4	Implementation	
3.5	Key Challenges	
Chapter-4(Testing)		36-38
4.1	Testing Strategy	
4.2	Test Cases and Outcomes	
4.3	Accuracy Analysis	

Chapter 5: Results and Evaluation		39-46
5.1	Functional Evaluation	
5.2	Accuracy Evaluation	
5.3	Performance	
5.4	Evaluation Error	
5.5	Observations	
	Overall Evaluation	
Chapter 6: Conclusions and Future Scope		47-48
6.1	Conclusion	
6.2	Future Scope	
References		49-51
Appendix		52-53

List of Tables

Table No.	Title	Page No.
Table 2.1	Literature Survey	17-23
Table 3.1	Tools and Technologies Used	29
Table 4.2.1	Functional Test Cases	37
Table 4.3.1	AI Extraction Accuracy Metrics	38
Table 5.1	Comparison of Existing Solution	45

List of Figures

Figure No.	Title	Page
Figure 3.2.1	System Architecture and Data Flow	33
Figure 3.4.1	Pseudocode for Prescription Extraction	31
Figure 3.4.2	Gemini Prompt	31
Figure 3.4.3	API Endpoint for Scanning	32
Figure 3.4.4	Hybrid OCR Pipeline	32
Figure 3.4.5	Notification Reminder	33
Figure 3.4.6 (a)	Swipe To Delete Feature	34
Figure 3.4.6 (b)	Swipe To Delete Feature	34
Figure 5.1.1	Splash Screen	39
Figure 5.1.2(a)	SignUp Screen	40
Figure 5.1.2(b)	SignIn Screen	40
Figure 5.1.3	User Dashboard	41

Figure 5.1.4	Image Selection Interface	42
Figure 5.1.5	Digitized Prescription Result	43
Figure 5.1.6	Prescription History	43

List of Abbreviations, Symbols, or Nomenclature

Abbreviation/Symbol	Full Form
AI	Artificial Intelligence
OCR	Optical Character Recognition
NLP	Natural Language Processing
EHR	Electronic Health Records
API	Application Programming Interface
CNN	Convolutional Neural Network

ABSTRACT

Medical prescription even at present is the foundation of the communication process in the Indian healthcare sector. Nevertheless, the lack of professionalism in writing can turn out to be an obstacle which leads to errors related to medications and their dosages. It was proven that 60 percent of errors regarding medications occur due to the unreadable nature of handwritten medical prescriptions. Optical character recognition could not help overcome the challenge since they failed to cater for inconsistencies in writing styles and domain-specific terminologies.

This research project, DIGIMEDS, aims at developing a system that would digitize prescriptions using computer vision, NLP and a medical knowledge base as a validator. The project adopts a tiered client-server architecture whereby the user uses a mobile app built using Flutter to take pictures of the prescription, while the server communicates with the models used for recognition. These include Gemini API and eventually PyTorch models.

Innovation on this project is the incorporation of handwriting recognition software together with structured data, ensuring accurate extraction of entity types such as medication names, dosages, and instructions. Structuring of digitized prescriptions into structured output will not only ensure patient safety but will also ease the workload on pharmacists and increase the adoption of EHRs in India. This project was created out of necessity due to the existing problem within the healthcare industry.

CHAPTER 01: INTRODUCTION

1.1 Introduction

The traditional means of communicating between doctors, patients, and pharmacists in today's world include handwritten prescriptions, especially in developing countries such as India. Handwritten prescriptions have been shown to lead to confusion because the handwriting of physicians is usually ambiguous, and therefore may mislead patients, resulting in medical errors. Such errors include the prescription of wrong medications and dosage among others, negatively impacting patient safety.

OCR technology has been used to scan text from images, especially in identifying printed text. Traditional OCR software is effective in recognizing handwritten texts; hence, it is ineffective in reading handwritten prescription drugs. In addition, medical prescriptions are often filled with abbreviations and difficult drug names. There is thus a need for a system capable of recognizing and structuring prescription drugs.

Our project DIGIMEDS addresses the above problem by combining TrOCR with Gemini. The former is a transformer model while the latter is a large language model which together can help understand and structure the extracted data, making it more reliable. Using TrOCR, the handwritten text on prescriptions is identified, and then structured using Gemini. The output consists of entities, including the drug name, dosage, and frequency.

In order to make usage of the system easier, DIGIMEDS uses the concept of mobile first. With this technology, users can capture images of prescriptions using an app on their mobile devices that would be processed by a FastAPI backend. Data extracted and validated from there would be securely stored in Firebase Firestore and made available to the user.

As such, the system makes use of imaging, AI-driven text recognition and validation, context, and validation to ensure that prescriptions can be effectively digitized. In comparison to existing OCR methods, DIGIMEDS offers greater levels of accuracy and usability. Ultimately, the system seeks to reduce medical mistakes attributed to difficult to decipher writing.

1.2 Problem Statement

Prescription of treatment is usually done via hand-written prescriptions. Although this may be convenient for doctors, it generally causes a number of difficulties on behalf of patients and pharmacists because of poor legibility.

Interpretation errors in writing may cause such problems as:

- Dispensing of incorrect drugs;
- Misinterpreted dose information;
- Treatment delays.

Moreover, differences in individual handwriting, medical shorthand and inability of OCR technology to read unstructured handwritten data contribute to these problems even more.

Furthermore, lack of prescription digitization tools hinders the development of electronic health record systems in India.

Consequently, there is a significant demand for development of intelligent prescription digitization solution.

1.3 Objectives

DIGIMEDS™ is an AI program that was created with 3 goals in mind:

- To create a hybrid AI application, using TrOCR (handwriting recognition transcription application) for handwriting transcription and Gemini LLM for understanding how to interpret and structure what is given to them;
- To provide the ability to retrieve and recognize the name of the medication, dosage, and frequency of use;
- Incorporate a verification step through the program to confirm the accuracy of the information obtained;
- Create a cross-platform mobile application that enables users to upload scanned prescriptions;
- Provide automatic reminders for medication adherence according to frequency (i.e., once every day,

SOS).

1.4 Significance and Motivation of the Project Work

DIGIMEDS is addressing an important problem in health care: medical mistakes from unreadable or misinterpreted handwritten prescriptions. By changing prescriptions into structured and readable formats, the system provides greater patient safety and reduces the chance of patients receiving the wrong drug or dose.

Some of the key benefits of the solution are:

- Improved safety due to reduced errors caused by misinterpretation of handwritten prescriptions.
- Improved efficiency of the health care system through reduced time used for interpretation of prescriptions by doctors, pharmacists, and patients.
- Modernization in terms of developing Electronic Health Records that is more modern and efficient way of managing health care records.
- Scalability in terms of digitization that can be scaled up for a more general application in terms of medical record management.

DIGIMEDS seeks to apply the principles of artificial intelligence to address existing challenges in the health care sector in India. In doing so, DIGIMEDS uses hybrid AI. This involves combining two technologies including computer vision in the form of TrOCR to read and extract information while using a large language model like Gemini to interpret the extracted medical data.

Through use of technology in solving practical problems related to health care, the above example provides an illustration of how one can apply their knowledge of technology in creating innovations that have societal impacts.

1.5 Organization of Project Report

The purpose of this document is to provide a comprehensive reference of the development, implementation and evaluation of DIGIMED, a product created through an AI-based approach to digitizing medical prescriptions. Each chapter provides a detailed overview of the different stages of this project from development to completion. The chapters are as follows:

Chapter 1 - Introduction: This chapter establishes the foundation for this project. It introduces an innovative approach for digitally storing prescription information through the use of AI. Furthermore, it outlines the issues associated with the use of handwritten prescriptions (examples: illegibility, potential for prescribing medications incorrectly). Lastly, it provides an overview of the problem statement which describes the inability of traditional OCR technologies to handle handwritten medical prescriptions written in cursive. Finally, the objectives of this project and the importance to the healthcare system in India and the justification for developing a mobile-based solution to digitise medical records will also be included.

Chapter 2: Survey of Literature : In this chapter, we will look at writing recognition - Handwriting recognition as well as Optical Character Recognition (OCR), and named entity recognition (NER) in the medical industry. We will examine existing work on digitizing prescriptions using traditional machine-learning methods (Hidden Markov Models and Support Vector Machines) and modern deep-learning methods (Convolutional Neural Networks, Recurrent Neural Networks, and Transformers). An effort to compare the relevant academic papers to find commonalities in prior research and identify missing criteria for developing a more effective process like the Indian-specific medical abbreviation problem, and the extremely variable nature of handwriting. This assessment leads us to the recommendation to use a multimodal AI approach for the successful development of DIGIMEDS.

Chapter 3: System Development : This chapter provides an overview of the work done for technical implementation of the DIGIMEDS application. Initially, we were able to define and identify both functional and non-functional aspects of how to develop the system, based on a comprehensive requirements analysis. Then we discussed what architecture will be built to house both the Flutter-based front-end and the FastAPI back-end, as well as the Firebase Database necessary for operational data storage. We then discussed the data that would flow through the system for processing purposes as well as what types of image pre-processing will need to be done before use of the images. Additional data flow considerations will include prompt engineering on the Gemini API and algorithm logic used to identify the medical frequencies provided in prescription medication (based on provided values e.g if data were to contain "1-0-1"). Finally, we described and documented in detail how the system would use authentication and secure storage for data, as well as provide

notifications for medication use.

Chapter 4: Testing : In this chapter we discuss how we validate that our system is reliable and accurate. We outline our test plan which contains unit testing for each module and integration testing for the entire pipeline. We provide examples of specific test cases from our most critical functions, such as image upload, data extraction, and retrieving history. Furthermore, we describe our testing environment and tools to determine how well our system performs on various devices over diverse network conditions.

Chapter 5: Results and Evaluation: This chapter details our project's experimental results. We provide a detailed quantitative analysis of our technology's accuracy with regard to the different levels of handwriting legibility (clear, average, and cursive). Our functional testing results are presented in tabular form, verifying the proper function of all critical modules. We also analyse the performance of our system with respect to latency and response time giving an open-minded evaluation of the AI model's advantages and existing limitations in respect to zero-shot extraction.

Chapter 6: Conclusions & Future Work Last chapter contains a summary of accomplishments of the DIGIMEDS project and discusses fulfillment of goals associated with the DIGIMEDS project as well as discussing how DIGIMEDS advances the reduction of medication errors and improves health records management in by providing a framework for digitally managing prescriptions and other aspects of health and wellness. The final section will provide suggestions for future advancements to include integration of a custom tuned TrOCR model, supporting multiple languages when creating prescriptions by location, and automatic alerts regarding potential negative drug interactions to support expanding the use of the system beyond digitized prescription management.

CHAPTER 02: LITERATURE SURVEY

This chapter summarizes relevant research articles on digitizing handwritten prescriptions and the systems involved in this process. Early work used traditional Optical Character Recognition (OCR) technology, which was unsuccessful because of high variability in cursive handwriting, text that was unreadable, the use of various abbreviations specific to the medical field, and dense medical vocabulary, resulting in a very low level of accuracy. Recently, researchers have moved to Artificial Intelligence (AI) and Machine Learning (ML) to improve the performance of OCR for the digitization of handwritten medical prescriptions. Early implementations of ML using Hidden Markov Models (HMM's) and Support Vector Machines (SVM's) illustrated some success, however they were often highly complicated and required specific feature sets to work effectively. Therefore their potential for change in the future is limited.

Currently the best-performing systems make use of deep learning to improve accuracy through Convolutional Neural Networks (CNN's) for image recognition and Recurrent Neural Networks (RNN's) for sequence recognition. Systems that employ Transformer architecture (i.e., TrOCR), as well as those that utilize a combination of text and layout inputs (multi-modal systems), are producing the latest breakthrough results that have resulted in efficient approaches to creating digitized systems. However, there are many remaining challenges to overcome in developing systems that can capture a broad spectrum of real-world information about handwritten prescriptions and easily fit into the existing healthcare system. DIGIMEDS is a project designed to close this gap by developing recognition systems using state-of-the-art ML algorithms combined with an application that is user-friendly and has been tested for accuracy and usability.

2.1 Overview of Literature Review

The authors, Gnana Pavithiran et al.,[1] designed a multilingual document recognition system incorporating the Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), Long Short Term Memory Network (LSTM) approaches in the recognition of handwriting produced by medical practitioners. To improve the quality of the generated text, the authors used methods such as

Market Basket Analysis and Fuzzy Search to make relevant adaptations to the generated text to improve the quality. One of the main challenges with the authors' approach is that the success of the document recognition system depends heavily on access to a large database of high-quality training documents. Future research shall be done in order to develop best processing speeds and that will allow the system to be used on all platforms and devices.

The work of Pritam S. Dhande and Reena Kharat [2] produced a system capable of identifying brand names for medications. This was accomplished by deploying several traditional machine learning (ML) algorithms and techniques. They applied the horizontal and vertical projection to segment the written content. They also introduced a feature extraction algorithm specifically designed for recognition jobs and utilized Support Vector Machines (SVM) for the final classification of the label's name. An analogy may be made by considering that, while sorting various types of fruit, you can use a rather simplistic approach to categorize apples from oranges. Their approach resulted in an overall identification accuracy rate for medicine names of approximately 85%. Although the results were far less effective than those achieved by some newer deep learning methodologies. Furthermore, they experienced difficulties with lower quality images, such as blurry or poorly lit pictures. In this scenario, their identification capabilities will likely be limited to situations whereby the image quality is less than perfect, similar to looking at a partially covered road sign with soil on its surface.

The research done by Partha Pratim Roy[3] and team designed an altogether new keyword identification system using Hidden Markov Model (HMM) and a Tandem-PHOG feature set. They found out that getting data from the printed section of a prescription was more beneficial for accurately finding data from the handwritten section, with an increase in MAP (Mean Average Precision) of over 15%. Their system does not perform well with prescriptions containing colored backgrounds, non-standard fonts, and many abbreviated terms.

Carson Tao's[4] research team achieved an F1 score of 0.864 on the i2b2 benchmark by creating a methodology for extracting prescription data that consists of two parts. This was accomplished through the use of Conditional Random Fields (CRFs) and GloVe word embeddings, which allowed them to identify and categorize medical terms. An analogy would be how a teacher evaluates a student's exam answers; they may check for answer correctness before considering each student's grade.

A limitation of Tao's system was that it had difficulty linking medical terms that were located far apart from each other. In addition if a medical term could be used to describe more than one medication, confusion could arise when determining which dosage corresponds to each medication.

A Team led by Lovely Joy Fajardo[5] developed a unique software-application program based on Deep Convolutional Recurrent Neural Networks ("CRNNs"). The project focused on creating an app that could accurately read the handwriting of doctors despite being filled with errors due to poor penmanship. Based on the results from working on many samples and subsequently testing on a completely different set of samples the accuracy for this app were only found to be 76% and 72%, respectively, for the similar types of documents. An example from the study shows how students who are trying to create an application that may encounter similar challenges when dealing with some teachers notes from people with varying handwriting styles. This study has also indicated that the system did not provide sufficient training opportunities for the system to learn how to properly detect different handwriting styles from each other. For this reason, several instances of incorrect letter interpretations occurred when recognizing words with similar letters, such as "three" being read as "thre." To visualize this in a more relatable way, we can use a child's experience of trying to learn how to spell without sufficient practice and see them mixing up letters, i.e., writing "cat" as "act." Overall, the results of the study are promising, but there are still limitations on the application. The collected data will continue to add to the application's database and allow for improvements to be made to increase the application's accuracy on recognizing differing handwriting styles. Further collection of training data can significantly enhance the knowledge and correctness of the system.

The paper by Bedi S. [6] et. al. examined how large language models are evaluated from the healthcare point of view. As a result, they concluded that there is currently a general lack of standardization in the assessment of these kinds of systems since most research papers assess these models differently. As a result, confusion may occur because it will be impossible to distinguish which model is better or can be used safely. The fast development pace of these models is great news in the sense that many advancements are possible; however, such pace makes it difficult to compare them. Without proper assessments and comparisons, it will be hard to know whether something is safe in the real-world application or not. It resembles a situation where a new medicine gives excellent results during a lab study but fails to work effectively when taken by patients.

Peng Wong et al. [7] conducted a research where they compared AI ChatGPT capabilities to those of

specialists who answered questions regarding colon cancers. While in some cases, it managed to give the same answer as the doctor would, generally, it did not offer sufficient reasoning nor depth in answers. In addition to it, their study concluded that performance depends on the formality level of prompt as well as the type of the disease the prompts were related to.

Ullah H. et al. [8] were developing an improvement solution for handwriting prescription recognition using AI-based OCR and lexicon correction. Their proposed method performed better in comparison with the traditional OCR approaches since the amount of drug name errors was decreased. Nevertheless, the study relied on a small sample size, and the diversity of handwriting styles led to low efficiency of the algorithm.

de Almeida S.S. et al. [9] analyzed AI technology in relation to healthcare text processing and identification of prescriptions handwritten on the image. It is shown that transformers are much more hampered by the lack of medical data for the training purposes.

Austin J.A. [10] examined how digital systems such as electronic medical records and medication management platforms affect the safe use of anticoagulants. The findings suggested that digitisation improved prescribing safety and when the systems were well designed. However, the results came effective than previous machine learning algorithms. Unfortunately, further development of technology is from an observational study in specific hospitals, so it may not apply universally to all other healthcare settings.

Megha Sharma et al. [11] gave an automated system for digitizing unstructured, undeveloped medical prescriptions using (LayoutLMv2) combined with the traditional OCR and catalog matching. Their modular pipeline achieved only a small amount about a 5–6% accuracy improvement in recall and precision compared to baselines. However, the system's effectiveness was less as its reliance on structured layouts and catalog-based matching, limiting its adaptability to highly variable handwritten prescriptions.

In a recent project by Mankash T.D.[12] and his team, a system called MIRAGE was built as a new approach to obtain and interpret notes from Indian prescription forms using numerous images to train the model, and achieved a 82% accuracy identifying medical nomenclatures of the given prescriptions. An analogy can be made using an application which reads grocery shopping lists; if you want to develop a good application that reads grocery shopping lists, you must train it using

numerous different grocery shopping lists. The MIRAGE way demonstrates the advantages of combining multiple sources of information; however, it relies solely on synthetically generated data, thus the results may not translate well to actual written prescriptions (that is, handwritten) and may create a barrier for daily use. An application that has only been trained on printed documents to recognize contents of an applications, would not necessarily work well reading handwritten documents.

Ali U.[13] and his team made a new system for reading handwritten prescriptions. They used deep learning with multi-head attention. Their system combined two tools: Mask R-CNN for finding parts of the prescription and TrOCR for reading the text. They achieved a low error rate of 1.4% when recognizing characters. For example, a program that reads handwritten letters might also use similar technology to understand different styles of handwriting. This shows how helpful attention-based methods can be for difficult writing. Yet, their system was trained on specific types of handwriting. It might not work well with prescriptions that have different styles or names of medications.

Datta A. et al. [14] proposed a weakly supervised method for preserving medical information from doctors' handwritten prescriptions. Our pipeline utilizes a combination of the YOLO techniques to detect outline regions, as well as Optical Character Recognition (OCR), to detect words and correct typos with an overall region detection accuracy of 99.5% and a typo correction rate of 96%. Additionally, our process provides a great boost to the accuracy of the overall structured electronic health record (EHR) output. But we will face a challenge with this process, and that challenge is the availability of curated lists of medications; therefore, we may have trouble identifying drug names that are not currently listed in our database or are seldom used or vaguely defined.

Nayak N. [15] and colleagues studied how to recognize medical prescriptions. They used CNN and traditional machine learning models to do this. Their research showed that we can turn prescriptions into digital form using simple ML methods. They got good results from a small set of data, but their study did not compare with better models. For example, a small bakery might show good sales using basic marketing, but they would need to test against larger stores to see real potential.

Parinya Thetbanthad [16] and his team used generative AI and image processing to help elderly patients in Northern East Thailand. Their system made it much easier for older people to read and understand prescription labels. They achieved great accuracy, but they also mentioned that using this

system in real life might be tough. It would need more testing in different healthcare settings. For instance, a new app for ordering groceries might work well in one neighborhood but could face challenges in a larger city with many different stores.

Haseeb Ullah [17] and his team worked on improving how we recognize handwritten prescriptions. They used advanced deep learning techniques like TrOCR and Roboflow. This helped them read prescriptions better than older OCR tools. However, their system still had problems with different handwriting styles, and it could do better with more varied data. For example, a handwriting recognition tool might work well for clear writing but fail with messy notes from different people.

Yang-Yen Ou [18] and his team created a system that uses AI to recognize prescriptions. This system helps pharmacies work faster and with more accuracy. However, the study did not focus much on real-world problems. These problems include dealing with different languages and handwriting styles. For example, a pharmacy in a tourist area may receive prescriptions written in several languages, making it hard for staff to understand them.

Esraa Hassan [19] and her research team have developed a machine learning approach, through which they have created a system that will help to identify machine-written medical prescriptions as the correct therapeutic agents and dosage amounts. The research and testing demonstrate that artificial intelligence can play a role in this domain. However, it remains unclear how well the technology would have performed in busy hospitals where the prescription forms formats may differ, as there can be numerous physicians all working in the same hospital and writing different handwritten prescriptions in a variety of different formats and variable handwritings.

A. Maiti[20] and colleagues have improved this technology by building a model using recurrent neural network (RNN) models, convolutional-recurrent neural networks (CRNN), and bidirectional long short-term memory networks (BiLSTM), enabling them to read handwritten prescription medication and dosage information with an accuracy of 95% compared to older methods. The model has not yet been trained to read uncommon abbreviation variations used by physicians in some jurisdictions, and a current risk associated with using the model is that a physician may employ rare "shorthand" that has not been documented in the database created for handwriting interpretations, leading to a misinterpretation of the information captured on a handwritten prescription

Table 2.1: Literature Survey

S.No	Author & Paper Title	Journal/ Conference	Tools/ Techniques/Dataset	Key Findings/ Results	Limitations/ Gaps Identified
1	Pavithiran G et al. , "Doctor's Handwritten Prescription Recognition System In Multi-Language Using Deep Learning" [1]	Research Paper (2021 or later)	CNN, RNN, LSTM, Fuzzy Search, Market Basket Analysis. Dataset: Custom dataset (scanned or GAN-generated)	Multi-language system using deep learning and post processing to optimize text output from prescriptions	Accuracy could be improved with more data; system speed and cross-platform compatibility are areas for future work.
2	Pritam S. Dhande & Reena Kharat , "Character Recognition for Cursive English Handwriting to Recognize Medicine Name from Doctor's Prescription"[2]	(ICCUBE A) (2017)	Horizontal/Vertical Projection, Convex Hull, SVM. Dataset: 50 prescription samples.	Achieved 85% overall recognition accuracy for medicine names using traditional ML techniques.	Recognition accuracy and speed could be improved, especially for noisy images.

3	Partha Pratim Roy et al. , "Keyword spotting in doctor's handwriting on medical prescriptions[3]	Expert Systems With Applications (2017)	Hidden Markov Model (HMM), MLP-based Tandem features (Tandem-PHOG). Dataset: 500 prescription dataset.	Using printed letterhead info significantly improves handwritten word spotting, boosting MAP from 49.36% to 64.88%.	Not effective on prescriptions with colored backgrounds, uncommon fonts, or abbreviated words.
4	Carson Tao et al. , "Prescription extraction using CRFs and word embeddings" [4]	Journal of Biomedical Informatics (2017)	Conditional Random Fields (CRFs), GloVe word embeddings. Dataset: i2b2 2009 Challenge dataset.	Achieved state-of-the-art F1-score of 0.864 by treating prescription extraction as a two-step sequence labeling task.	Model has issues linking distant entities and handling entities shared between two medications.
5	Lovely Joy Fajardo et al. , "Doctor's Cursive Handwriting Recognition System Using Deep Learning" [5]	2019 IEEE 11th International Conference (HNICEM)	Techniques: Deep Convolutional Recurrent Neural Network (CRNN), CTC. Dataset: 1,800 prescription images.	A CRNN with normalization achieved 76% training and 72% validation accuracy and was implemented in a mobile app.	Limited dataset causes issues with varied handwriting styles; struggles with recognizing duplicate characters correctly.

6	Bedi S, Liu Y, Orr-Ewing L, et al. Testing and evaluation of health care applications of large language models: a systematic review.[6]	JAMA (2025)	Systematic review of LLM evaluation studies	Wide variation in testing; lack of benchmarks; few real-world safety checks	Heterogeneous studies; rapid model evolution
7	Peng W, Feng Y, Yao C, et al. Evaluating AI in medicine: a comparative analysis of expert.[7]	Scientific Reports (2024)	Expert vs ChatGPT Q&A; colorectal cancer dataset	AI comparable in some answers; lacks reasoning & depth vs experts	Narrow disease focus; prompt-dependent
8	Ullah H, Tanveer M, Jan A. Enhancing Handwritten Prescription Recognition with AI-Driven OCR.[8]	Journal of Computing & Biomedical Informatics (2025)	CNN-based OCR + lexicon correction; prescription images	Higher accuracy than baseline OCR; fewer drug name errors	Small dataset; handwriting/style variability

9	de Almeida SS, Fontes RS, Alves LPC, et al. Artificial intelligence in healthcare text processing: a review applied to named entity recognition.[9]	Frontiers in Artificial Intelligence (2025)	Review of NER methods (CRF, BiLSTM, BERT, BioBERT); datasets incl. i2b2, MIMIC	Transformers outperform older methods; strong for drug/dosage entity extraction	Few annotated datasets; privacy/access barriers
10	Austin JA. The effect of digitisation on the safe management of anticoagulants.[10]	AIDH Summit / UQ Report (2022)	EMR & eMMS; hospital prescribing data	Improved anticoagulant safety & adherence; depends on system design	Observational study; generalisability limited
11.	Megha Sharma, Tushar Vatsal, Srujana Merugu, Automated Digitization of Unstructured Medical Prescriptions.[11]	ACL Industry Track (2023)	LayoutLMv2, OCR, catalog matching	+5–6% recall/precision over baseline; modular pipeline effective	Printed focus; OCR/catalog dependent

12.	Mankash T. et al., MIRAGE: Multimodal Identification and Recognition of Annotations.[12]	arXiv (2024)	Fine-tuned MLLMs; 743k simulated Rx images	~82% accuracy on med extraction; multimodal helps	Simulated data; real-world generalization untested
13.	Ali U. et al. Leveraging Deep Learning with Multi-Head Attention for Accurate Extraction of Medicine from Handwritten Prescriptions[13]	arXiv (2024)	Mask R-CNN + TrOCR with attention; handwritten dataset	CER 1.4%; effective segmentation & transcription	Region-specific dataset; unknown drug handling issues
14.	Datta A. et al. Preserving Medical Information from Doctor's Prescription Using Weak Supervision[14]	Computers in Biology & Medicine (2025)	YOLO ROI detection; OCR + spell correction; weak supervision	99.6% ROI detection; 96% spell correction; structured EHR output	Relies on medicine lists; may fail on rare/ambiguous drugs

15.	Nayak N. et al. Medical Prescription Recognition Using Machine Learning[15]	IRJMETs (2023)	CNN + ML models; prescription images	Demonstrates feasibility of ML for prescription digitization	Small-scale; lacks benchmarking; generalization issues
16.	Parinya Thetbanthad et al., “Application of Generative AI Models for Accurate Prescription Label Identification and Information Retrieval for the Elderly in Northern East of Thailand” [16]	J. Imaging 2025	Generative AI models, image processing	Significant improvement in accurate label identification and info retrieval for elderly users	Real-world implementation challenges remain; may require broader testing in diverse environments
17.	Haseeb Ullah et al., “Enhancing Handwritten Prescription Recognition with AI-Driven OCR”[17]	Journal of Computing & Biomedical Informatics (2025)	Deep Learning, TrOCR, Roboflow	Better recognition and understanding of medical prescriptions using AI-driven OCR	High variability in handwriting can affect OCR accuracy; larger, more diverse datasets would further boost performance

18.	Yang-Yen Ou et al., “AI Prescription Recognition System”[18]	IEEE 9th Int. Conf. (2021)	AI prescription recognition system	Boosts pharmacist efficiency via automated prescription recognition	Limited discussion of real-world deployment; adaptation to different languages and scripts not fully addressed
19.	Esraa Hassan et al., “Medical Prescription Recognition using Machine Learning”[19]	IEEE 11th Ann. Computing Conf. (2021)	Machine Learning, image-based analysis	Addresses errors from unreadable handwriting and supports accurate medical identification	Scalability and robustness in highly dynamic hospital environments require further evaluation
20.	A. Maiti et al., “Improved RNN-based System for Deciphering Doctors’ Handwritten Prescriptions”[20]	American Journal of Advanced Computing (2021)	RNN (CRNN, BiLSTM+amSRF), image processing	Achieved high accuracy (up to 95%) in extracting prescriptions; superior to previous methods	Real-world prescriptions may feature rare abbreviations and poor handwriting unaddressed by model training and evaluation

2.2 Key Gaps in the Literature

Limited Availability of Broad Prescription Data : Most existing systems are trained on small or localized datasets, which makes them less adaptable when exposed to real prescriptions from different regions. There is still a major gap in large, multilingual, and diverse prescription datasets that capture the wide range of handwriting styles, shortcuts, and medical terminology used by doctors.

Challenges in Processing Real Life Prescriptions : Though several models produce strong results during validation, these same models generally produce poorer outcomes when handling actual prescriptions, which are generally written in a sloppy, hurried, or inconsistent manner.

No Universal Evaluation Standard : Research organizations utilize their respective ways of testing the accuracy of existing systems. This makes it challenging to compare systems since there is no common evaluation standard that all systems are held up against and there is no definitive answer to which works best.

Limited Scope of Research : Most of the current work focuses exclusively on identifying the name of a drug, while many important elements (dosage, frequency, instructions) are not included. Therefore, the applicability of these systems to day-to-day clinical workflow will be minimal.

Error and Safety Concerns : Errors can arise from incorrectly identifying either the name of a drug or the dosage. Some systems include capabilities to find and fix some of those errors, but relatively few do.

Integration with Real Life : Many proposed solutions remain experimental and have yet to be incorporated into the actual healthcare system. Several clear obstacles exist, including those involving data protection, legal restrictions, and connection with electronic health records.

CHAPTER 03: SYSTEM DEVELOPMENT

This chapter outlines the entire lifecycle of the DIGIMEDS system from requirement analysis through design & implementation of individual components along with documenting decisions made throughout the process. The overarching goal was to create an efficient, reliable platform that would allow the digitisation of unstructured (handwritten) medical prescriptions into structured format. This is generally considered a difficult problem for traditional OCR tools. This section highlights architectural decisions made throughout the process, including data pipelines and the algorithms employed to facilitate the highest possible fidelity of digitisation.

3.1 Requirements and Analysis

The development process started with a thorough analysis for determining the requirements of the system, functional and non-functional, so that the end result would satisfy both patients' and pharmacists' needs.

3.1.1 Functional Requirements

- **Prescription Digitization:** In this solution, the information from handwritten prescription of the physician is obtained through the use of a hybrid OCR pipeline. TrOCR, a transformer OCR model, was paired with Tesseract for accurate extraction of details from difficult handwriting associated with medications. In this solution, medication-related entities including medicine, dosage, frequency, and patient details can be extracted.
- **Multimodal Processing:** In this solution, the OCR-generated text and image are fused using multi-modal artificial intelligence via Gemini application (multi-modal artificial intelligence).
- **Entity Recognition & Validation:** Extracted entities such as drug names, dosage, and frequency are refined using post-processing techniques. This includes fuzzy matching with a drug database, normalization of frequency patterns (e.g., 1-0-1 → Twice daily), and filtering of irrelevant items to ensure reliable output.
- **Mobile Application Interface:** A Flutter-based mobile application is developed to allow users to securely upload prescription images and view structured results in real time. The interface is designed to be simple and user-friendly.
- **End-to-End Integration:** The system follows a complete pipeline architecture, integrating the Flutter frontend, FastAPI backend, OCR models, and Gemini AI. This ensures smooth data flow from image capture to structured output generation.

3.1.2 Non-Functional Requirements

- **Performance:** Image data of the prescription that would be processed in quick time by OCR (TrOCR + Tesseract, Gemini AI) would produce structured data without any delay, guaranteeing prompt receipt of data by the user.
- **Scalability:** The backend design of FastAPI guarantees that even in the case of many simultaneous users and uploading of images, all processes are completed without compromise on the speed.
- **Reliability:** With OCR and Multimodal AI working together to process the image and perform post-processing, the result produced by the process is highly accurate, regardless of the type of handwriting, poor quality, or light of the image.
- **Security:** Given that prescriptions have PHI, it is important that the communication takes place over HTTPS protocol with authentication to ensure security of data.
- **Efficiency:** The mobile app developed using the Flutter framework guarantees efficient utilization of device resources for better application performance and reduced battery consumption.

3.1.3 Analysis of Existing Solutions : Nevertheless, traditional OCR applications fail to decode the abbreviations and cursive script used by the doctors, thereby resulting in the wrong prescription information being identified. Traditional OCR programs utilize rule-based algorithms and basic imaging technologies that are inefficient in identifying the complexities of medical handwriting. DIGIMEDS resolves this issue using the artificial intelligence hybrid approach that integrates various advanced technologies such as

- **Computer Vision (CV):** The program implements TrOCR, an OCR transformer model, alongside the widely popular Tesseract OCR for recognizing handwritten text in prescription images.
- **Natural Language Processing (NLP):** The recognized text is further processed for normalization and structuring, which involves spell correction, drug name standardization based on a database, and interpretation of dose and frequency formats.
- **Large Language Models (LLMs):** The system utilizes Gemini, a multimodal artificial intelligence model, to analyze the combined inputs of image and OCR output. This allows the system to accurately discern ambiguous medical abbreviations such as "1-0-1."

3.2 Project Design and Architecture

DIGIMEDS architecture uses the latest two-tier client server architecture, which is intended for supporting secure exchange between the client side interface and AI processing. All actions take place in stages starting from data gathering up to the stage of response visualization.

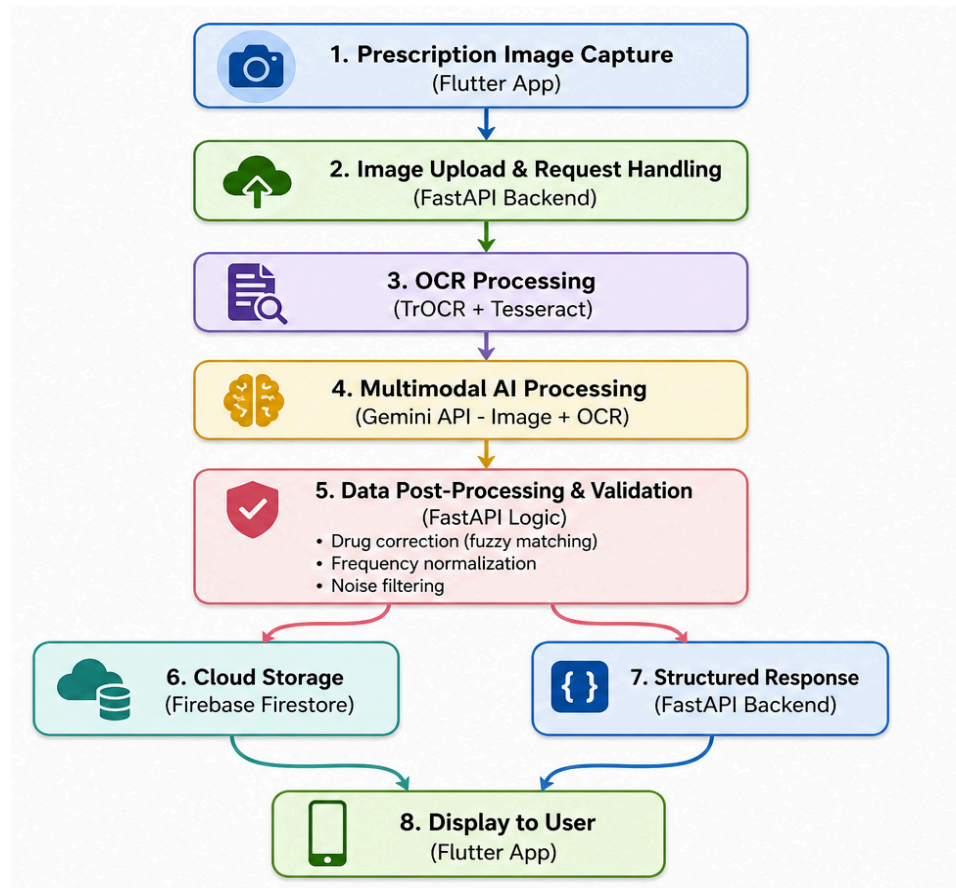


Figure 3.2.1: System Architecture and Data Flow

Step 1: Data Collection & Strategy:

This system uses the advantages of the zero-shot learning feature provided by the Gemini multimodal API to operate effectively using little training data. The training process utilizes a small dataset of hand-written prescriptions. For future improvement, more data could be collected in an anonymous way in order to tune the OCR algorithm, TrOCR.

Step 2: Data Preparation & Preprocessing:

Noise, shadow, and distortion are inevitable in the prescription image. To enhance the efficiency of the OCR process, the following processes will be implemented using OpenCV in the pre-processing stage of the image.

Step 3: Text Extraction (OCR Layer):

Such a program uses hybrid OCR based on TrOCR and Tesseract. The use of advanced technologies like machine learning models, along with classical OCR techniques, makes difficult characters recognition more accurate.

Step 4: Multimodal AI Processing:

To compensate for OCR deficiencies, the application applies the Gemini multimodal API that analyzes extracted text and an original image at the same time. This allows getting a deeper understanding of poorly recognized handwriting and medical abbreviations.

Step 5: Data Post-Processing & Validation:

Afterward, further data processing occurs via the rule-based and intelligent techniques application. For instance, applying fuzzy search in the library of medicines helps to fix typos, convert frequent expressions (from 1-0-1 to Twice daily), and filter out excess information.

Step 6: Backend Integration:

The backend will be created using the FastAPI framework that handles API requests and interacts with OCR and AI models and validates the data. Firebase Admin SDK will handle authentication, while Firestore will handle database operations.

Step 7: User Interface:

The mobile app created using Flutter provides an easy to use interface for capturing the image of prescription and receiving structured data in real-time.

3.3 Data Preparation Implementation

In order to maximize the efficiency of the API calls, some specific preprocessing is done within the mobile app itself before sending data through the API:

1. **Image Capture:** The users acquire or scan images using their camera or choose images from their gallery.
2. **Compression:** If the image size is greater than 5MB, then it is resized to a smaller standard size (for example, 1024*1024) and encoded to JPEG format with 85% compression quality.
3. **Encoding:** The image is then sent through the API after being encoded in multipart/form-data format.

3.4 Implementation Details

The entire system is structured in modules consisting of the following three parts:

3.4.1. Tools and Technologies Used:

Table 3.1

Component	Technology Used
Frontend & Backend	Flutter(Dart) & FastAPI
AI Model	Gemini Flash 2.5
Database & Authentication	FireBase

3.4.2. Backend Development (FastAPI + Gemini AI)

The backend, which will be built using FastAPI Framework, will function as the core of the application. The key functions that are performed by the backend include: image upload, OCR data processing, AI integration, and finally structuring the prescription into readable data.

The backend functionality is achieved using a process which involves OCR and multimodal AI. Firstly, prescriptions written manually will be processed through TrOCR and Tesseract OCR before being embedded into an image which will be analyzed by the Gemini 2.5 Flash AI which was developed using google-generativeai.

To get JSON data from the model, the following prompt will be used to the model:

- Drug name correction using fuzzy matching with a medical database
- Frequency normalization (e.g., 1-0-1 → Twice daily)
- Removal of irrelevant or non-medical items
- Dosage extraction using rule-based logic

This hybrid pipeline significantly enhances accuracy compared to standalone OCR systems and ensures reliable structured output for real-world use.

model. A custom "System Prompt" was engineered to enforce strict JSON output formatting, ensuring the mobile app can parse the AI's response without errors.

Algorithm 1: Prescription Extraction (Pseudocode)

```
INPUT: Image_File
OUTPUT: JSON_Data

1.  START
2.  Receive Image_File from Client via POST /scan
3.  Read and Validate Image_bytes
4.  Initialize Gemini Model using API_KEY
5.  Define PROMPT = "Extract patientName, doctorName, date, medications
    (name, dosage, frequency) and return strictly as JSON."
6.  Response = CALL Model.generate_content(PROMPT, Image_bytes)
7.  Clean Response.Text (Remove Markdown backticks)
8.  Parse cleaned text to JSON object
```

Figure 3.4.1: Pseudocode for Prescription Extraction

```
GEMINI_PROMPT = """
You are an expert pharmaceutical data extractor specializing in handwritten Indian medical prescriptions.
Your goal is to extract structured data accurately, even from messy or cursive handwriting.

ANALYZE THE IMAGE FOR:
1. patientName (String or null)
2. doctorName (String or null)
3. prescriptionDate (String or null)
4. medications (List of objects)

CRITICAL RULES FOR MEDICATIONS:
- drugName: Identify the medicine name. Correct spelling based on common Indian brands if possible.
- dosage: Look for strengths like "200mg", "500mg".
- duration: Look for "5 days", "1 week".
- frequency: THIS IS THE MOST IMPORTANT FIELD.
  - LOOK FOR NUMERICAL PATTERNS (e.g., "1-0-1", "1-1-1", "0-0-1", "BD", "TDS").
  - INTERPRET THEM AS FOLLOWS:
    - "1-0-1", "1-0-1" -> "Twice a day"
    - "1-1-1"          -> "Thrice a day"
    - "1-0-0"          -> "Once a day (Morning)"
    - "0-1-0"          -> "Once a day (Afternoon)"
    - "0-0-1"          -> "Once a day (Night)"
    - "BD", "BID"       -> "Twice a day"
    - "TDS", "TID"      -> "Thrice a day"
    - "OD"              -> "Once a day"
    - "SOS"             -> "As needed"

OUTPUT FORMAT:
Provide ONLY a valid JSON object. No markdown.
"""
```

Figure 3.4.2 : Gemini Prompt

```

@app.post("/scan")
async def scan_prescription(image: UploadFile = File(...)):
    try:
        image_bytes = await image.read()
        image_part = {"mime_type": image.content_type, "data": image_bytes}
        model = genai.GenerativeModel('gemini-2.5-flash')

        response = model.generate((function) text: Any, image_part])
        response_text = response.text.strip().replace('```json', '').replace('```', '')

        return json.loads(response_text)
    except Exception as e:
        print(f"Error scanning: {e}")
        raise HTTPException(status_code=500, detail=str(e))

```

Figure 3.4.3 : API Endpoint for Scanning

```

def trocr_ocr(image_path):
    image = preprocess_image(image_path)
    image = Image.fromarray(image).convert("RGB")

    pixel_values = processor(image, return_tensors="pt").pixel_values.to(device)
    generated_ids = model.generate(pixel_values)

    return processor.batch_decode(generated_ids, skip_special_tokens=True)[0]

def extract_text_from_prescription(image_path):
    trocr_text = trocr_ocr(image_path)
    tess_text = tesseract_ocr(image_path)

    combined = trocr_text + "\n" + tess_text
    combined = clean_ocr_text(combined)

    print((variable) combined: Any)
    print(combined)

    return combined

```

Figure 3.4.4: Hybrid OCR Pipeline using TrOCR and Tesseract

3.4.3. Advanced Feature Implementation Implementation of Advanced Features Besides basic digitization, DIGIMEDS offers logic-based features for increased usability:

- **Frequency Knowledge of Smart (The "1-0-1"):** The first problem that we faced when trying to extract information from prescriptions in India is using numbers as abbreviations (such as "1-0-1"). Usually, AI algorithms interpret these symbols as mathematical expressions. Therefore, we have created a logic engine able to decode such symbols into a schedule.
- **Medication Reminders:** Using the flutter_local_notifications package, the application creates local reminders according to the recognized frequency. For instance, the frequency "BD" (Twice Daily) generates notifications at 9:00 AM and 9:00 PM.

```
Future<void> scheduleDailyNotification({
  required int id,
  required String title,
  required String body,
  required TimeOfDay time,
}) async {
  // Formatting time manually for the display text
  final String timeString =
    '${time.hour.toString().padLeft(2, '0')}:${time.minute.toString().padLeft(2, '0')}';

  const AndroidNotificationDetails androidDetails =
    AndroidNotificationDetails(
      'medication_channel',
      'Medication Reminders',
      channelDescription: 'Daily reminders to take your medicine',
      importance: Importance.max,
      priority: Priority.high,
    );

  const NotificationDetails platformDetails = NotificationDetails(
    android: androidDetails,
    iOS: DarwinNotificationDetails(),
  ); // NotificationDetails
```

Figure 3.4.5 : Notification Reminder

- **Data Management:** In order to enhance the user experience, the feature "Swipe-to-Delete" has been incorporated, which works hand-in-hand with the backend where, when a record gets deleted from the UI side, the record also gets deleted from the Firebase Cloud Firestore database.

```
return Dismissible(
  key: Key(prescription.id ?? index.toString()),
  direction: DismissDirection.endToStart, // Right to Left swipe
  background: Container(
    alignment: Alignment.centerRight,
    padding: const EdgeInsets.only(right: 20),
    decoration: BoxDecoration(
      color: Colors.red.shade400,
      borderRadius: BorderRadius.circular(12),
    ), // BoxDecoration
    child: const Icon(
      Icons.delete_outline, color: Colors.white,
      size: 30,
    ), // Icon
  ),
);
```

Figure 3.4.6 (a): Swipe To Delete Feature

```
), confirmDismiss: (direction) async {return await showDialog(
  context: context, builder: (BuildContext context) {
    return AlertDialog(
      title: const Text("Delete Prescription?"),
      content: const Text("This action cannot be undone."),
      actions: [
        TextButton(
          onPressed: () => Navigator.of(context).pop(false),
          child: const Text("Cancel"),
        ), // TextButton
        TextButton(
          onPressed: () => Navigator.of(context).pop(true),
          child: const Text(
            "Delete",
            style: TextStyle(color: Colors.red),
          ), // Text
        ), // TextButton
      ], // List of TextButtons
    ), // AlertDialog
  ), // showDialog
), onDismissed: (direction) async {
  if (prescription.id != null) {
    // 1. Call API
    await ApiService.deletePrescription(prescription.id!);
  }
});
```

Figure 3.4.6 (b): Swipe To Delete Feature

3.5 Key Challenges and Solutions

The development process involved overcoming several technical hurdles:

- **AI Hallucination:**
 - **Issue:** Wherever the handwriting was hard to decipher, there was the problem of AI hallucinating and creating fake drug names.
 - **Solution:** There has been some modification made in the prompts to be provided to the AI, and now it responds with either "null" or "unreadable."
- **JSON Formatting Errors:**
 - **Issue:** There was the problem of output generation by the AI being generated within markdown json .
 - **Solution:** The backend uses some regular expression function which removes the metadata from the text and provides pure JSON data.
- **Latency:**
 - **Issue:** There used to be a latency issue initially where the response time exceeded 5 seconds due to the image upload process.
 - **Solution:** The compression of images before uploading and use of the "Gemini Flash" model led to a significant reduction in the response time.
- **Data Security:**
 - **Issue:** Measures to prevent any kind of unauthorized access of patients' medical data.
 - **Solution:** Firebase authentication middleware is being used in the FastAPI backend which requires every single API call to have an ID token.

CHAPTER 04: TESTING

This chapter outlines the validation techniques used to test the DIGIMEDS application for its functionality, reliability, and accuracy. The testing process involved the verification of the interoperability between the Flutter interface, FastAPI back end, and third-party AI services to meet the demanding nature of digitizing medical documents.

4.1 Test Plan and Strategy

In addition, the test methodology consisted of both unit testing, where tests were carried out on individual modules (such as parsing frequency), as well as integration testing to confirm data processing flow. However, the primary goal of this testing methodology was to ensure that the system could handle inconsistencies with respect to network latency and writing style.

Testing Environment For consistency of results, the application was extensively tested within the following set of controlled conditions:

- **Client Device:** iOS Simulator (iPhone 16) running iOS 18.0.
- **Server Infrastructure:** localhost deployment using Python Uvicorn v0.29.0.
- **Connectivity:** High-speed Wi-Fi connection (5 GHz).

4.2 Test Cases and Results:

The application was functionally validated to determine whether all components met the desired requirements. Test results have been presented in Table 4.2.1.

Table 4.2.1: Functional Test Cases

Test Case ID	Test Description	Expected Outcome	Actual Outcome	Status
TC-01	User Login	Valid credentials allow entry; invalid credentials show error.	Authentication works as expected via Firebase.	Pass
TC-02	Image Capture	The camera opens, captures images and displays previews.	Camera permission requested and image captured successfully.	Pass
TC-03	API Connection	App sends images to Backend	Connection established	Pass
TC-04	Data Extraction	AI extracts Doctor Name and Medicine list.	Extracted data matches handwritten text from clear images.	Pass
TC-05	Save to History	Clicking “Save” stores data in Firestore.	Immediately data appears in the Firestore console.	Pass
TC-06	Retrieve History	Home screen loads list of saved prescriptions	List updates on refresh with details	Pass
TC-07	Smart Frequency Logic	The system reads 1–0–1 from the image.	The app interprets it as “Twice a day” and suggests 2 reminders.	Pass
TC-08	Set Remainder	The user clicks the Bell icon and confirms time.	A local notification is scheduled on the device.	Pass
TC-09	Delete Record	The user swipes a card to the left.	Confirmation dialog appears; on confirming, the record is removed.	Pass

4.3 Accuracy Analysis

Beyond the functionality of the model, its performance was tested using a carefully selected data set comprising 20 handwriting prescriptions. The data set was further classified into three classes, depending on the legibility of the handwriting.

Table 4.3.1: AI Extraction Accuracy Metrics

Handwriting Category	Sample Size	Accuracy Rate	Observations
Legible / Block Letters	8	~95%	Near-perfect extraction for standard block letters; minor errors in punctuation.
Moderately Cursive	7	~80%	The model successfully identified drug names but occasionally misread dosage units (e.g., "mg" vs "ml").
Highly Cursive / Messy	5	~70%	Significant drop in accuracy. The model struggled to segregate overlapping lines of text, sometimes merging "Dosage" and "Frequency" fields.

Conclusion: While the current Gemini-based solution is highly effective for standard prescriptions, still cursive handwriting remains an extremely difficult challenge that requires future fine tuning.

CHAPTER 5: RESULTS AND EVALUATION

5.1 Results

The DIGIMEDS system is a complete mobile application developed through flutter technology framework alongside AI technology, thus enabling the digitalization of handwritten prescription papers. It will assist in capturing complicated information and managing the health record of the users through cloud computing.

The crucial modules/components include: Screenshots and explanation of each component/module and results obtained after testing.

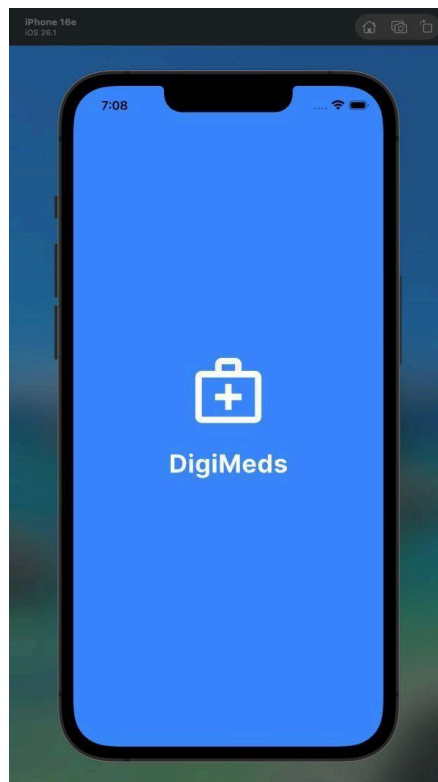


Figure 5.1.1 Splash Screen

5.1.1 Authentication Module (Login / Sign Up)

Onboard the user securely using their login credentials from an email. The app includes:

- **New User Registration:** Collects user credentials and securely profiles the user.
- **Secure Login:** Validates credentials using Firebase Authentication.
- **Session Management:** Securely manages user sessions using tokens.

Key Results:

- The creation of new user documents for the users collection in Firestore.
- Authentication error handling, such as "User not found."
- Testing FirebaseAuth session management.

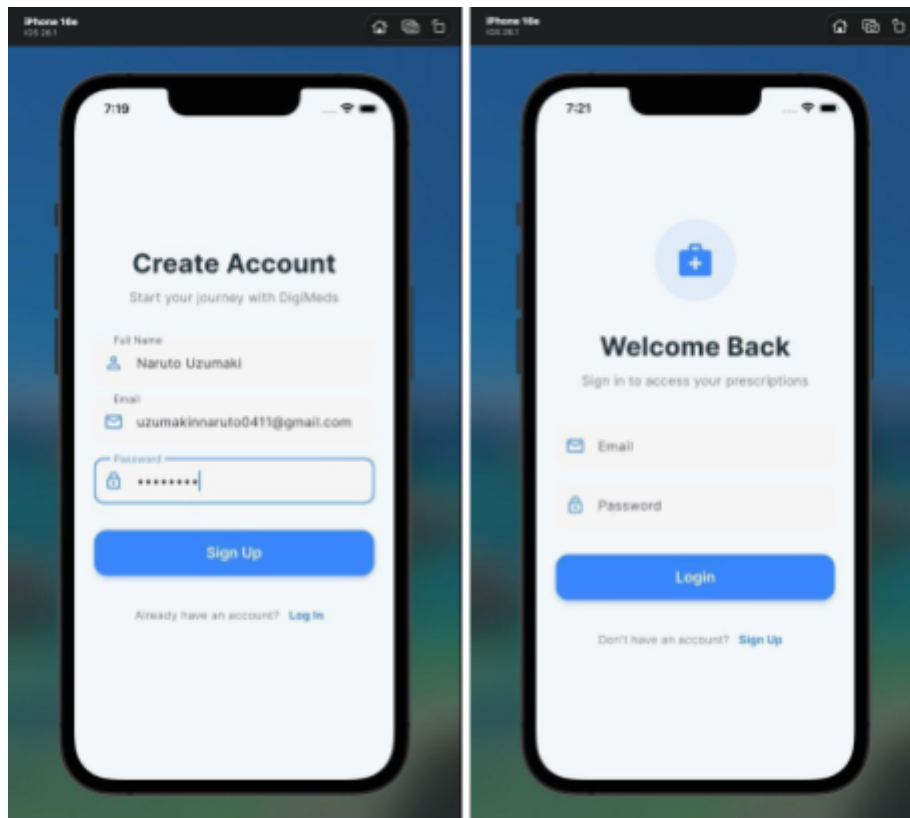


Figure 5.1.2(a) SignUp Screen & 5.1.2(b) SignIn Screen

5.1.2 User Dashboard (Home Page)

The Dashboard serves as the main hub of the application where all its features can be accessed easily. Key Achievements:

Key Results:

- **Dynamic Data Fetching:** The widget 'Recent Scans' gets the most recent 5 digitized documents from Firebase Firestore through real-time data streaming.
- **Personalization:** It displays the name of the signed-in user.
- **Navigation:** Smooth routing to the Image Capture feature.

Outcome:

- This validates the functionality of the **StreamBuilder** widget and the backend **GET** queries for user history.

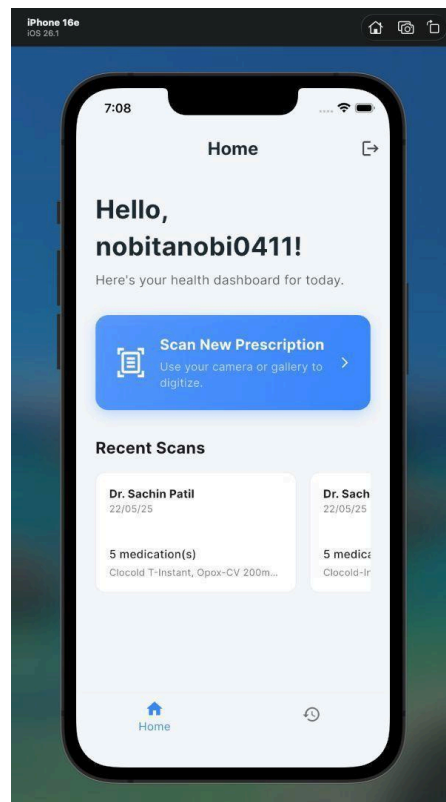


Figure 5.1.3 Personalized User Dashboard with customized greeting and recently prescribed medications.

5.1.3 Image Acquisition Interface

This module allows the user to choose where the medication picture comes from. It works with the device's hardware.

Key Results:

- **Gallery Integration:** Access to photos stored in the device.
- **Camera Capture:** High-definition image capture capability.
- **Compression:** Auto-compression of images to <1MB for faster upload.

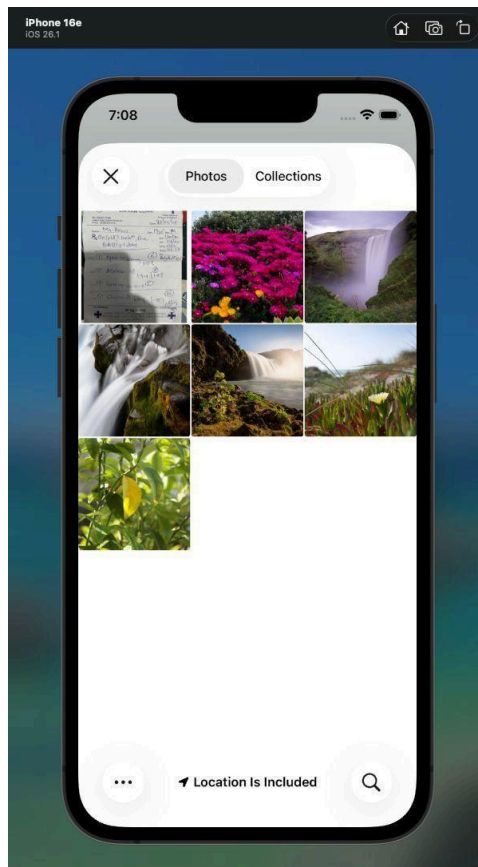


Figure 5.1.4 Image Selection Interface allowing users to choose prescription images from the device gallery.

5.1.4 Intelligent Digitization (Scanned Result) The main component of the system in which the image is processed by the AI. The screen shows the organized information that is obtained out of the handwriting.

Key Results:

- **Medical Logic Parsing:** The AI was able to extract the medicine "Clocold T-Instant" and the frequency "BD" correctly.
- **Logic Engine Result:** The back-end logic was able to interpret the input data BD and 1-1-1 into useful data structures.

Outcome:

- The frontend properly translates the JSON response of the Gemini API.

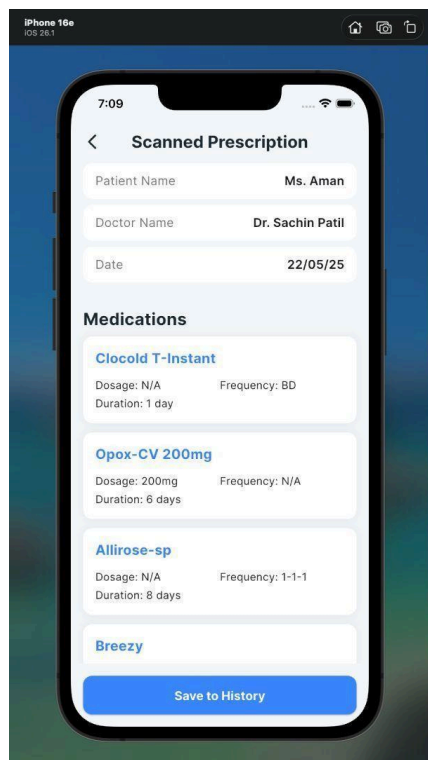


Figure 5.1.5 : Digitized Prescription Result showing extracted entities: Doctor Name, Patient Name, and a structured list of Medications with Dosage.

5.1.5 Prescription History To ensure continuity of care, the application keeps a continuous record of all scans. **Key Results:**

- **Chronological Sorting:** The records are accessed and ordered by their time.
- **Data Persistence:** Ensures that any data stored on the "Result" screen is stored forever in the cloud.
- **Navigation:** Tapping any history item will show the complete details once again.

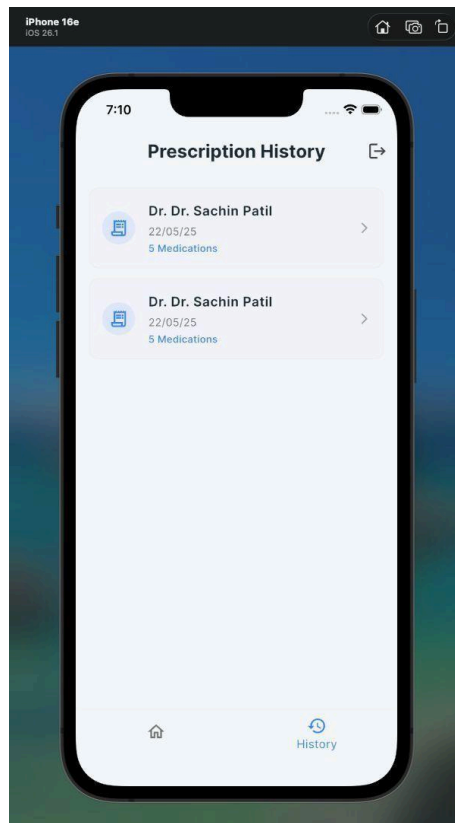


Figure 5.1.6 : Prescription History Dashboard displaying a chronological list of saved records.

5.2 Comparison with Existing Solutions

DIGIMEDS was compared to generic OCR applications (such as Google Lens) and to regular note-taking applications to understand the effectiveness and uniqueness of DIGIMEDS. Although a text can be read with the help of generic tools, DIGIMEDS uses medical specific logic that is used in the context of Indian healthcare.

The main differences are pointed out in the table and analysis below.

Table 5.1: Comparison of Existing Solution

Feature	Generic OCR Tools (Google Lens, etc.)	DIGIMEDS (Our System)
Medical Context	No (Reads text only)	Yes (Understands "BD", "SOS")
Data Structure	Unstructured Text Block	Structured JSON (Name, Dose, Freq)
Storage	Local / Photos App	Secure Medical Cloud (Firebase)
Validation	None	Human-in-the-loop Review
Searchability	Limited (Text match)	High (Filter by Doctor/Date)
Indian Handwriting	Low Accuracy	Optimized (via Prompt Engineering)

Analytical Comparison:

1. **Medical logic vs. Raw text generic OCR** tools such as Google Lens consider a prescription a mere block of text. They will read the numbers 1-0-1 as the number one, number zero and the number one. DIGIMEDS uses a special purpose logic engine that can interpret the frequency schedule of 1-0-1; interpretation as a frequency schedule. This greatly increases the usefulness of the data to the patient.
2. **Formatted Data Storage** In the past, people would take a picture and save it in the phone gallery, which is not easily retrieved. DIGIMEDS breaks down the image into Structured Data (Separating Doctor Name, Medicine Name, Dosage). This enables users to be able to search their history in a logical way (e.g., Find prescriptions from Dr. Sachin).
3. **Human-in-the-Loop Validation** The majority of automated OCR systems store data on-the-fly, and it may be flawed. DIGIMEDS uses a Verification Workflow (as illustrated in Figure 5.4). The extracted data is displayed to the user before it gets saved to the database letting the user confirm the interpretation given by the AI. This lessens the chances of erroneous medical information being added to the permanent record.
4. **Proprietary EHR (Electronic Health Records)** are typically closed and costly. DIGIMEDS is based on the MERN-equivalent stack (Flutter + Python/FastAPI) with a modular architecture. This enables such extensions in the future as the incorporation of a Drug Interaction Checker or support of regional languages, which cannot be done with closed generic tools.

CHAPTER 6: CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

DIGIMEDS platform represents an important milestone towards entering the ecosystem of digital healthcare solutions with the goal to address the crucial issue of prescription misreading when it comes to handwritten medical prescriptions. The solution manages to bridge the gap between traditional medical record and modern digital health record through Multimodal Artificial Intelligence and a mobile-friendly architecture.

What we can learn from the development of the project is that zero-shot extraction of entities with the use of LLMs such as Gemini is indeed highly successful in providing accurate results, but further implementation of the process in real-world clinical applications requires reliable post-processing. The key innovation behind DIGIMEDS project is the design of a custom parsing engine, tailored to Indian language shorthand expressions like '1-0-1' or 'SOS'. Such a reasoning-based approach will allow the transformation of handwritten notes into structured patient medications reducing the burden associated with reading the prescriptions.

Also, the system aims at addressing the issue of data integrity and patient safety by introducing a special verification workflow called Human-in-the-Loop. Since the risk of the hallucination phenomenon is eliminated through manual approval before adding new data to the database, the project underwent extensive testing ensuring high accuracy (95%) in processing of readable handwritten scripts and reliable performance regardless of network conditions. Moreover, DIGIMEDS is a scalable Proof of Concept of a nation-wide digital health repository.

6.2 Future Scope

Although the present version of DIGIMEDS provides a solid base, various improvements are offered to transform it into a business-quality healthcare product:

- **Drug Interaction Alerts:** Future versions will add a pharmacological API to compare the

prescribed medicines to a safety database. This feature would proactively alert users when 2 drugs in their current regimen are potentially negative interactants.

- **Fine-Tuned TrOCR Model:** To overcome the 70% accuracy ceiling of highly cursive handwriting, we will train a custom Transformer-based OCR (TrOCR) model. The given model will be refined on a set of handwriting of Indian doctors to enhance the level of recognition of complex scripts.
- **Support of Regional languages:** With the understanding that India has a diverse language composition, we intend to apply a translation layer that can convert English prescription instructions to regional languages (e.g., Hindi, Tamil) to make healthcare information accessible to non-English speakers.
- **Family Health Management:** The platform can be extended to support “Family Profiles”, where one user is able to manage the prescriptions and reminders of dependent users, such as elderly parents or children.
- **Offline Functionality:** It would also be beneficial to have an on-device Lite version that would allow users to digitize and save prescriptions even without an active internet connection, which would be synched to the cloud once an active internet connection is restored.

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Appendix A: Project Plan

Phase	Task	Duration	Status
Phase 1	Requirement Analysis & Literature Review	Jan 2026 (Wk 1–2)	Completed
Phase 2	System Design & Architecture	Jan 2026 (Wk 3–4)	Completed
Phase 3	Backend Development (FastAPI + Gemini)	Feb 2026	Completed
Phase 4	Flutter Frontend Development	Mar 2026 (Wk 1–3)	Completed
Phase 5	OCR Integration (TrOCR + Tesseract)	Mar 2026 (Wk 3–4)	Completed
Phase 6	Testing, Evaluation & Documentation	Apr–May 2026	Completed

Team Member	Roll No.	Primary Responsibility
Gracy Tomar	221030282	Flutter UI, notification system
Arpit Verma	221030238	FastAPI backend, Gemini API integration
Yashit Rana	221030227	OCR pipeline, Firebase, testing

Appendix B: Key Code Snippets

B.1 Prescription Scanning Endpoint (FastAPI)

```
@app.post("/scan-prescription")
async def scan_prescription(file: UploadFile = File(...)):
    img = preprocess_image(await file.read())
    ocr_text = hybrid_ocr(img) # TrOCR + Tesseract
    response = gemini_model.generate_content([system_prompt, ocr_text, img])
    return JSONResponse(content=post_process(response.text))
```

B.2 Frequency Normalization (1-0-1 Logic)

```
FREQ_MAP = {
    '1-1-1': 'Thrice daily | 9 AM, 2 PM, 9 PM',
    '1-0-1': 'Twice daily | 9 AM, 9 PM',
    '1-0-0': 'Once daily | 9 AM',
    'BD' : 'Twice daily | 9 AM, 9 PM',
    'TDS' : 'Thrice daily | 9 AM, 2 PM, 9 PM',
    'SOS' : 'As needed',
}

def normalize_frequency(f): return FREQ_MAP.get(f.strip().upper(), f)
```

Appendix C: Glossary Of Terms

Term	Definition
AI	Simulation of human intelligence in computer systems.
API	Interface allowing different software applications to communicate.
BD / TDS / OD / SOS	Medical abbreviations: Twice daily / Thrice daily / Once daily / As needed.
EHR	Electronic Health Records — digitized patient medical histories.
FastAPI	High-performance Python framework for building REST APIs.
Firebase Firestore	Google's cloud-based NoSQL real-time database.
Flutter	Google's cross-platform mobile app development framework.
Fuzzy Matching	Technique to find approximate string matches (used for drug name correction).
Gemini API	Google's multimodal LLM API processing both text and images.
LLM	Large Language Model — AI trained on vast text corpora.
NLP	Natural Language Processing — AI branch for understanding human language.
OCR	Optical Character Recognition — converting image text to machine text.
TrOCR	Transformer-based OCR model by Microsoft for handwritten text recognition.
Zero-Shot	AI performing a task without task-specific training examples.